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D-MEF2: A MADS box transcription factor expressed in differentiating mesoderm and muscle cell lineages during *Drosophila* embryogenesis

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ABSTRACT **The myocyte enhancer factor (MEF) 2 family of transcription factors has been implicated in the regulation of muscle transcription in vertebrates. We have cloned a protein from *Drosophila*, termed D-MEF2, that shares extensive amino acid homology with the MADS (*M*CM1, *A*gamous, *D*eficiens, and *s*erum-response factor) domains of the vertebrate MEF2 proteins. *D-mef2* gene expression is first detected during *Drosophila* embryogenesis within mesodermal precursor cells prior to specification of the somatic and visceral muscle lineages. Expression of *D-mef2* is dependent on the mesodermal determinants twist and snail but independent of the homeobox-containing gene tinman, which is required for visceral muscle and heart development. *D-mef2* expression precedes that of the *MyoD* homologue, nautilus, and, in contrast to nautilus, *D-mef2* appears to be expressed in all somatic and visceral muscle cell precursors. Its temporal and spatial expression patterns suggest that *D-mef2* may play an important role in commitment of mesoderm to myogenic lineages.**

The formation of skeletal muscle during embryogenesis involves the commitment of mesodermal progenitors to the myogenic lineage followed by the expression of muscle structural genes. The muscle-specific basic–helix–loop–helix proteins, MyoD, myogenin, myf5, and MRF4, have been shown to regulate muscle development (reviewed in ref. 1), but little is known of the mechanisms that control these regulators during the earliest steps of myogenic lineage specification.

In *Drosophila*, the mesoderm-specific regulatory genes twist and snail are responsible for the formation of mesoderm at the beginning of gastrulation (reviewed in ref. 2). Later, the mesoderm separates into the somatic muscle lineage, which gives rise to skeletal muscle, and the visceral muscle lineage, which gives rise to muscles of the gut and heart. Because the separation of these two muscle lineages occurs after the expression of twist and snail, but well before the expression of the MyoD homologue nautilus in somatic muscle cells (3, 4), it is likely that one or more regulatory genes act in this intervening period to specify the somatic and visceral muscle lineages. One such gene appears to be the homeobox-containing gene tinman, which is first expressed in the uncommitted mesoderm and subsequently becomes restricted to visceral mesoderm and heart (5, 6).

Members of the myocyte enhancer factor (MEF) 2 family of regulatory factors have been implicated in the control of *myogenin* and *MyoD* gene expression in vertebrates and are therefore potential candidates for early regulators of the skeletal muscle lineage (7–12). MEF2 was originally defined as a muscle-specific DNA-binding activity that recognizes a conserved A+T-rich element associated with numerous muscle-specific genes (13). The recent cloning of *MEF2* has revealed that it belongs to the MADS (*M*CM1, *A*gamous, *D*eficiens, and *s*erum-response factor) family of transcription factors. Four *MEF2* genes, designated *MEF2A*, *-B*, *-C*, and *-D*, have been cloned from vertebrates (10, 14–19). The proteins encoded by these genes are highly homologous within the 55-amino-acid MADS domain at their amino termini and within an adjacent MEF2-specific region of 27 residues, but they diverge outside of these regions.

We have cloned a *Drosophila* homologue of MEF2, termed D-MEF2,† which, to our knowledge, is the first MADS protein to be identified in *Drosophila*. During *Drosophila* embryogenesis, *D-mef2* is expressed first in the committed mesoderm and subsequently in the somatic and visceral muscle lineages. Its temporal and spatial patterns of expression suggest that *D-mef2* functions at an early step in a genetic cascade leading to the specification of myogenic lineages in the *Drosophila* embryo.

MATERIALS AND METHODS

Drosophila Stocks. Stocks carrying the sna^{IIG05} and twi^{ID96} mutant alleles were obtained from the Bloomington stock center. The tin^{EC40} mutant allele was generously provided by R. Bodmer (University of Michigan) (5). The tinman strain contained a marked TM3 balancer chromosome (TM3-Pw^+-*lacZ*), which permitted the identification of homozygous tinman mutant embryos by their lack of *lacZ* expression.

***D-mef2* cDNA Isolation.** A *Drosophila* 3- to 12-hr embryonic cDNA library in λgt10 (kindly provided by Tom Kornberg, University of California, San Francisco) was screened at low stringency (17) with a mouse *MEF2A* probe containing the MADS/MEF2 domain (J. Martin and E.N.O., unpublished results).

***In Situ* Hybridization to Whole-Mount Embryos.** *In situ* hybridization of a digoxigenin-labeled *D-mef2* DNA to wild-type and mutant embryos was as described (20).

***In Vitro* Transcription and Translation and Gel Mobility Shift Assays.** *In vitro* transcription and translation were carried out using the TNT rabbit reticulocyte lysate *in vitro* translation system (Promega) according to manufacturer's instructions. A *D-mef2* cDNA, encompassing the complete open reading frame (nucleotides +535 to +2295), was cloned into the P-CITE vector (Novagene). Gel mobility shift assays were performed as described (17) using an oligonucleotide encompassing the muscle creatine kinase (MCK) MEF2 site (13) as probe. Competitor oligonucleotides were added at 100-fold molar excess to the labeled probe. Sequences of

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Abbreviations: MEF, myocyte enhancer factor; MADS, *M*CM1, *A*gamous, *D*eficiens, and *s*erum-response factor; MCK, muscle creatine kinase; CAT, chloramphenicol acetyltransferase; tk, thymidine kinase.
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†The *D-mef2* cDNA sequence reported in this paper has been deposited in the GenBank data base (accession no. U03292).

oligonucleotides were as follows: MCK MEF2, GATCGCTCTAAAAATAACCCTGTCG; mutant 6, GATCGCTCTAAACATAACCCTGTCG.

Transfections. Transient transfections of S2 cells were carried out using a calcium phosphate precipitation method. Chloramphenicol acetyltransferase (CAT) activity was measured as described (13). The constructs used were 4xMEF2-tkCAT, which contains four copies of the MCK MEF2 site upstream of the thymidine kinase (tk) promoter linked to CAT, and the parental vector tkCAT, which lacks the MEF2 sites. The *D-mef2* expression vector contained nucleotides +1 to +1939 of the cDNA under control of the Moloney sarcoma virus (MSV) promoter in the vector EMSV.

RESULTS

Isolation of cDNA Clones Encoding D-MEF2. To search for potential *Drosophila* MEF2 homologues, a *Drosophila* 3- to 12-hr embryonic cDNA library was screened under conditions of reduced stringency with a portion of a mouse *MEF2A* cDNA encompassing the MADS box and MEF2 domain. The longest positive clone was 1900 bp in length and encoded an open reading frame that began with an ATG codon immediately preceding a region with striking homology to the MADS and MEF2 domains of the mammalian MEF2 proteins. Thus, we designated the protein encoded by this cDNA D-MEF2, for *Drosophila* myocyte-specific enhancer factor 2, and we designated the gene encoding this protein *D-mef2*. The complete open reading frame of D-MEF2, deduced from the composite sequences of multiple overlapping cDNA clones, encodes a 515-amino-acid protein with a predicted M_r = 54,289 (pI = 8.2) (Fig. 1*A*). The *D-mef2* cDNA hybridizes to a single transcript of ≈3.5 kb in embryos and adults (data not shown).

Within the MADS domain, D-MEF2 differs from the four mammalian MEF2 proteins at only 5–7 of 56 residues (Fig. 1*B*). The MEF2 domain of D-MEF2 is also highly conserved over the first 21 residues, but it diverges from the vertebrate

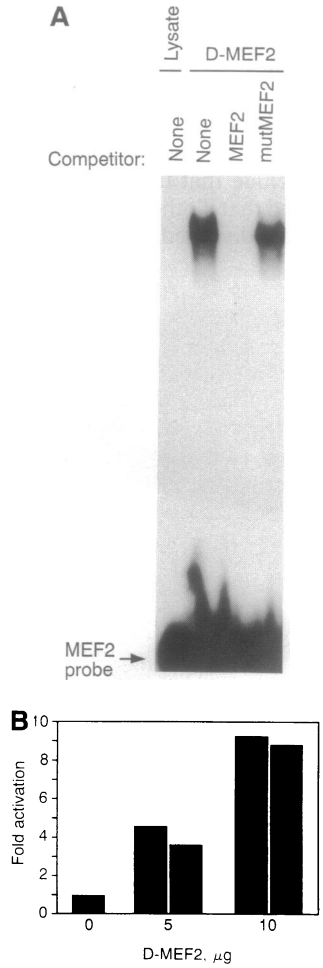

FIG. 2. D-MEF2 binds to and activates transcription through the MEF2 site. (*A*) An oligonucleotide corresponding to the MEF2 site from the mouse MCK enhancer was used as a probe in a gel mobility shift assay with *in vitro*-translated D-MEF2. Unprogrammed lysate was included in a parallel lane. A 100-fold excess of unlabeled MEF2 or mutant MEF2 oligonucleotide was used as competitor. (*B*) Schneider cells were transiently transfected with 5 μg of the 4xMEF2-tkCAT or tkCAT reporter genes and 5 or 10 μg of the EMSV expression vector with and without the *D-mef2* cDNA insert. CAT activity was determined as described in the text. Values are expressed relative to the basal level of expression of tkCAT, which was set at 1. The results of duplicate transfections are shown.

A

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ATGGGCCGCAAAAAAATTCAAATATCACGCATCACCGATGAACGCAATCGGCAGGTGACCTTCAACAAGCGCAAGTTCGGCGTGATGAAGAAGGCCTACGAGCTGTCCGTGCTCTGCGAC
M   G   R   K   K   I   Q   I   S   R   I   T   D   E   R   N   R   Q   V   T   F   N   K   R   K   F   G   V   M   K   K   A   Y   E   L   S   V   L   C   D    40
TGCGAGATCGCCCTGATCATCTTCTCGTCGAGCAACAAGCTGTACCAGTACGCCAGCACCGACATGGATCGCGTCCTGCTCAAGTACACCGAGTACAACGAGCCCCACGAGTCCCTCACC
C   E   I   A   L   I   I   F   S   S   S   N   K   L   Y   Q   Y   A   S   T   D   M   D   R   V   L   L   K   Y   T   E   Y   N   E   P   H   E   S   L   T    80
AACAAGAACATCATCGAGAAGGAGAACAAGAACGGCGTGATGTCGCCGGACTCGCCCGAAGCCGAAACGGACTACACACTCACTCCGCGAACGGAGGCCAAGTACAACAAGATCGACGAG
N   K   N   I   I   E   K   E   N   K   N   G   V   M   S   P   D   S   P   E   A   E   T   D   Y   T   L   T   P   R   T   E   A   K   Y   N   K   I   D   E   120
GAGTTCCAGAACATGATGCAGCGCAACCAGATGGCCATCGGCGGTGCGGGTGCCCCTCGCCAGCTTCCAAACAGCAGCTACACGCTGCCCGTTTCTGTTCCGGTGCCGGGATCTTACGGC
E   F   Q   N   M   M   Q   R   N   Q   M   A   I   G   G   A   G   A   P   R   Q   L   P   N   S   S   Y   T   L   P   V   S   V   P   V   P   G   S   Y   G   160
GACAACCTGCTGCAGGCCAGTCCACAGATGTCCCACACCAACATCAGCCCCCGTCCATCGAGTTCGGAGACGGATTCAGGTGGGATGTCCCTGATAATCTATCCATCGGGTTCCATGCTG
D   N   L   L   Q   A   S   P   Q   M   S   H   T   N   I   S   P   R   P   S   S   S   E   T   D   S   G   G   M   S   L   I   I   Y   P   S   G   S   M   L   200
GAGATGTCGAACGGCTATCCGCATTCACACTCGCCGCTTGTGGGATCACCGAGTCCGGGTCCCAGTCCTGGCATAGCCCACCATTTGTCCATTAAGCAGCAGTCGCCGGGCAGCCAGAAC
E   M   S   N   G   Y   P   H   S   H   S   P   L   V   G   S   P   S   P   G   P   S   P   G   I   A   H   H   L   S   I   K   Q   Q   S   P   G   S   Q   N   240
GGACGAGCTTCCAACCTAAGGGTCGTCATACCGCCCACAATTGCCCCCATACCGCCCAATATGTCAGCGCCGGATGATGTGGGATATGCAGATCAACGACAGAGCCAGACATCGCTTAAC
G   R   A   S   N   L   R   V   V   I   P   P   T   I   A   P   I   P   P   N   M   S   A   P   D   D   V   G   Y   A   D   Q   R   Q   S   Q   T   S   L   N   280
ACGCCAGTGGTCACGCTGCAGACGCCGATTCCCGCCCTCACGAGCTATTCCTTTGGGGCGCAGGACTTTTCCTCCTCCGGCGTAATGAACAGCGCGGATATCATGAGCCTCAACACCTGG
T   P   V   V   T   L   Q   T   P   I   P   A   L   T   S   Y   S   F   G   A   Q   D   F   S   S   S   G   V   M   N   S   A   D   I   M   S   L   N   T   W   320
CATCAGGGCCTGGTGCCGCACTCTAGTCTCTCGCACCTGGCTGTCTCGAATAGCACGCCGCCGCCCGCCACCTCCCCCGTCTCCATAAAGGTCAAGGCTGAGCCGCAGTCGCCGCCCGAGA
H   Q   G   L   V   P   H   S   S   L   S   H   L   A   V   S   N   S   T   P   P   P   A   T   S   P   V   S   I   K   V   K   A   E   P   Q   S   P   P   R   360
GATCTTTCGCCAGCGGTCATCAGCAGAATAGCAATGGTTCCACGGGCAGGCGGCGGATCCAGCAGCAGCACCAGTAGCAACGCCAGCGGAGGAGCAGGAGGCGGTGGAGCCGTCAGCGCA
D   L   S   P   A   V   I   S   R   I   A   M   V   P   R   A   G   G   G   S   S   S   S   T   S   S   N   A   S   G   G   A   G   G   G   G   A   V   S   A   400
GCCAATGTCATCACGCACTTGAACAACGTCAGTGTCCTGGCGGGAGGTCCTTCGGGGCAGGGAGGAGGAGGCGGAGGCGGCGGCAGCAACGGGAATGTCGAACAGGCCACCAATCTTAGC
A   N   V   I   T   H   L   N   N   V   S   V   L   A   G   G   P   S   G   Q   G   G   G   G   G   G   G   G   S   N   G   N   V   E   Q   A   T   N   L   S   440
GTACTGAGCCACGCGCAGCAACATCACCTGGGCATGCCCAACTCGCGTCCCTCGTCCACGGGCCACATCACACCCACTCCAGGTGCGCCCGAGCAGCGACCAGGATGTGCGTCTGGCAGCC
V   L   S   H   A   Q   Q   H   H   L   G   M   P   N   S   R   P   S   S   T   G   H   I   T   P   T   P   G   A   P   S   S   D   Q   D   V   R   L   A   A   480
GTCGCCGTGCAGCAGCAACAACAGCAGCCACATCAGCAACAGCAACTAGGCGACTACGATGCCCCCAACCACAAACGGCCGAGAATATCGGGCGGATGGGGCACATAG
V   A   V   Q   Q   Q   Q   Q   Q   P   H   Q   Q   Q   Q   L   G   D   Y   D   A   P   N   H   K   R   P   R   I   S   G   G   W   G   T   *       516
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B

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                  MADS BOX                                                MEF2 DOMAIN
D-MEF2    1  MGRKKIQISRITDERNRQVTFNKRKFGVMKKAYELSVLCDCEIALIIFSSSNKLYQYASTDMDRVLLKYTEYNEPHESLTNKNIIE  86
mMEF-2A   1  MGRKKIQITRIMDERNRQVTFTKRKFGLMKKAYELSVLCDCEIALIIFNSSNKLFQYASTDMDKVLLKYTEYNEPHESRTNSDIVE  86
hMEF-2B   1  MGRKKIQISRILDQRNRQVTFTKRKFGLMKKAYELSVLCDCEIALIIFNSANRLFQYASTDMDRVLLKYTEYSEPHESRTNTDILE  86
mMEF-2C   1  MGRKKIQITRIMDERNRQVTFTKRKFGLMKKAYELSVLCDCEIALIIFNSTNKLFQYASTDMDKVLLKYTEYNEPHESRTNSDIVE  86
mMEF-2D   1  MGRKKIQIQRITDERNRQVTFTKRKFGLMKKAYELSVLCDCEIALIIFNHSNKLFQYASTDMDKVLLKYTEYNEPHESRTNADIIE  86
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FIG. 1. Nucleotide sequence and deduced open reading frame of D-MEF2. (*A*) The nucleotide sequence of the coding region and deduced reading frame of D-MEF2 are shown. (*B*) Homology of D-MEF2 to human (h) and mouse (m) MEF2 proteins. Sources of sequences are mMEF2A, J. Martin and E.N.O., unpublished results; hMEF2B, ref. 14; mMEF2C, ref. 17; mMEF2D, ref. 18.

proteins in the last 8 residues. Outside of the MADS box and MEF2 domain, D-MEF2 shows no significant homology to previously described MEF2 proteins.

Southern hybridization analysis of *Drosophila* genomic DNA with a *D-mef2* cDNA probe under low-stringency conditions yielded bands that could all be attributed to *D-mef2* (data not shown). We conclude that there is only a single *D-mef2* gene in the genome and that if other MADS box genes exist in *Drosophila*, they have diverged considerably from *D-mef2*. By *in situ* hybridization of the *D-mef2* cDNA to polytene chromosomes obtained from the salivary glands of third instar larvae, *D-mef2* was mapped to the right arm of chromosome 2 in the 46C interval (data not shown).

D-MEF2 Binds the Same DNA Sequence as Its Vertebrate Homologues. When D-MEF2 was translated *in vitro* and was tested for its ability to bind a ^{32}P-labeled oligonucleotide corresponding to the MEF2 site from the mouse MCK enhancer, a prominent DNA–protein complex was observed

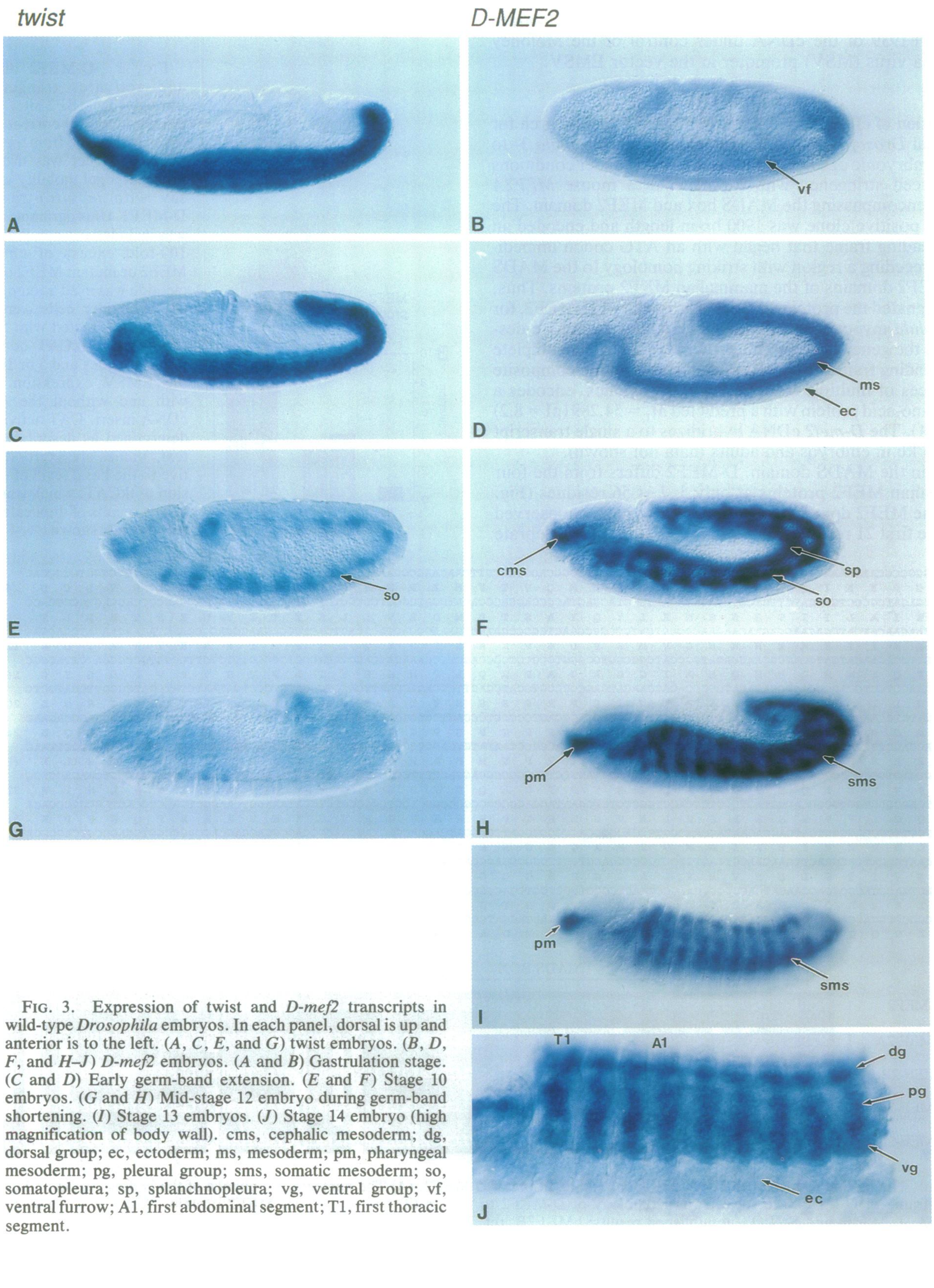

FIG. 3. Expression of twist and *D-mef2* transcripts in wild-type *Drosophila* embryos. In each panel, dorsal is up and anterior is to the left. (*A*, *C*, *E*, and *G*) twist embryos. (*B*, *D*, *F*, and *H–J*) *D-mef2* embryos. (*A* and *B*) Gastrulation stage. (*C* and *D*) Early germ-band extension. (*E* and *F*) Stage 10 embryos. (*G* and *H*) Mid-stage 12 embryo during germ-band shortening. (*I*) Stage 13 embryos. (*J*) Stage 14 embryo (high magnification of body wall). cms, cephalic mesoderm; dg, dorsal group; ec, ectoderm; ms, mesoderm; pm, pharyngeal mesoderm; pg, pleural group; sms, somatic mesoderm; so, somatopleura; sp, splanchnopleura; vg, ventral group; vf, ventral furrow; A1, first abdominal segment; T1, first thoracic segment.

(Fig. 2*A*). DNA binding by D-MEF2 was sequence-specific and was competed by the cognate site but not by a mutant site (mutant 6) that does not bind the vertebrate MEF2 proteins.

To determine whether D-MEF2 could function as a transcriptional activator, we placed the *D-mef2* cDNA under control of the Moloney sarcoma virus promoter and transfected this expression vector into *Drosophila* S2 cells with a CAT reporter linked to the tk basal promoter alone or with four copies of the MCK MEF2 site. D-MEF2 efficiently activated transcription of only the MEF2-dependent reporter (Fig. 2*B*), indicating that transactivation required direct interaction of D-MEF2 with its target sequence.

D-mef2 Transcripts Are Localized to Mesodermal Precursor Cells That Give Rise to Somatic and Visceral Muscle. *D-mef2* transcripts were localized in developing embryos by *in situ* hybridization, and their expression pattern was compared to that of twist, an early marker for mesoderm formation (2). *D-mef2* gene expression is first detected in cells of the ventral furrow during gastrulation (Fig. 3*B*). The spatial pattern of *D-mef2* expression at this stage is similar to that of twist (Fig. 3*A*), but the level of *D-mef2* RNA accumulation is lower and the initial expression of *D-mef2* does not extend to the anterior pole as does twist. During germ-band extension (Fig. 3 *C* and *D*), transcripts for twist and *D-mef2* accumulate in an identical pattern and are restricted to the mesoderm cell layer. In early stage 10 embryos, *D-mef2* RNA is detected only in the single layer of mesoderm and faintly detected in the cells that will give rise to the pharyngeal muscles (data not shown), an expression pattern comparable to twist at this stage (Fig. 3*E*). In late stage 10, the mesoderm, initially organized as an epithelium, separates into two different cell layers. The inner splanchnopleura gives rise to visceral musculature and heart precursor cells while the outer somatopleural layer of mesoderm gives rise to the somatic musculature (2). *D-mef2* expression is high in both layers of the mesoderm, and transcripts are also seen at high levels in precursors of the pharyngeal musculature (Fig. 3*F*). At this stage it is clear that transcripts are present in both layers of the mesoderm in all thoracic and abdominal segments as well as the gnathal segments. During germ-band retraction, the *D-mef2* RNA localization pattern undergoes a series of refinements. As the germ-band retracts (Fig. 3 *H* and *I*), *D-mef2* transcript levels begin to decline in the visceral musculature and heart precursor cells but remain at high levels in the cells of the somatic musculature. In contrast, twist RNA accumulation is barely detectable by this stage (Fig. 3*G*). In stage 13 embryos, *D-mef2* transcripts are still present in the segmental clusters of the somatic musculature, and expression is maintained in these cells as dorsal closure continues (Fig. 3*J*). In the later stages of embryogenesis, *D-mef2* continues to be expressed in cells of the somatic musculature and the dorsal vessel but not in other mesoderm derivatives including the fat body.

The D-mef2 Expression Pattern Is Disrupted in twist and snail Mutant Embryos. To begin to define the position of the *D-mef2* gene within the genetic hierarchy leading to myogenesis, we examined *D-mef2* gene expression in twist and snail mutant embryos, which fail to form mesoderm. In stage 11–12 mutant embryos, there was little or no *D-mef2* expression detected (Fig. 4), suggesting that the *D-mef2* gene lies in a regulatory pathway downstream of these mesodermal regulators. Tinman is expressed in the mesoderm concomitant with *D-mef2* at germ-band extension and is required for the formation of visceral mesoderm (5, 6). *D-mef2* was expressed normally during germ-band extension in tinman mutant embryos, indicating that *D-mef2* is not regulated by tinman in the uncommitted mesoderm.

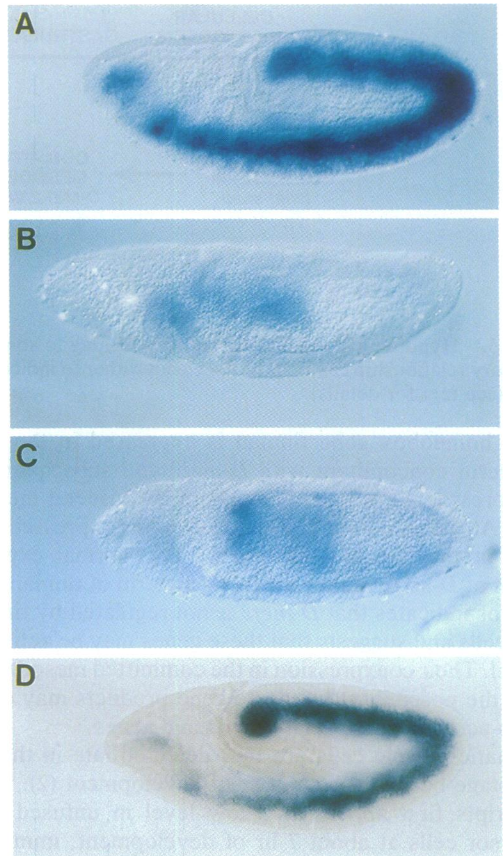

FIG. 4. Expression of *D-mef2* transcripts in twist, snail, and tinman mutant embryos. Whole mounts of wild-type embryos (*A*) or embryos homozygous for the twi^{ID96} (*B*), sna^{IIG05} (*C*), or tin^{EC40} (*D*) alleles were examined for *D-mef2* expression. (*C*) In snail mutants during germ-band extension, a very low level of *D-mef2* expression was observed. Later-stage snail mutants showed no expression above background. Staining in the posterior midgut of the mutant and wild-type embryos represented nonspecific background.

DISCUSSION

Mesoderm formation during *Drosophila* embryogenesis is controlled by twist and snail and begins at gastrulation when cells in the midventral region of the embryo invaginate to form a ventral furrow (2). Embryos lacking twist or snail develop normally until gastrulation, but they fail to form the ventral furrow and form no mesodermal derivatives (2). *D-mef2* expression is activated at gastrulation, following the initial expression of twist, and is found in all twist-expressing cells by mid-germ-band extension. The close temporal and spatial correlation of *D-mef2* and twist expression, combined with the observation that *D-mef2* expression is disrupted in twist mutant embryos, raises the possibility that the *D-mef2* gene may be a direct target for transcriptional activation by the twist protein. Likewise, the substantial reduction of *D-mef2* RNA accumulation in snail mutant embryos implicates the snail gene product as a direct or indirect regulator of *D-mef2*.

After invagination of the presumptive mesoderm within the ventral furrow, the more ventral mesodermal cells become committed to the somatic muscle lineage, whereas the more dorsal mesoderm gives rise to the visceral muscles of the gut and heart (2). Expression of *D-mef2* is first detected in the germ-band extended embryo within mesodermal precursors of the somatic and visceral muscle lineages. During germ-band retraction, *D-mef2* expression declines in the visceral mesoderm, but its expression increases to high levels in the somatic mesoderm.

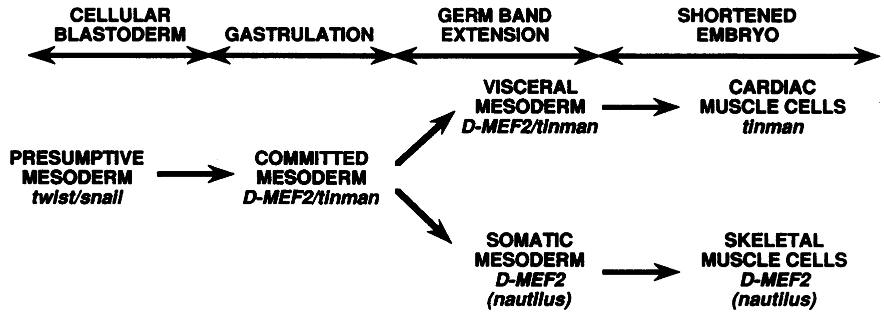

FIG. 5. Hypothetical regulatory pathway leading to somatic and visceral muscle in *Drosophila*. The arrows are not meant to imply direct regulatory relationships among the genes but rather to indicate their expression patterns within the myogenic lineage at different developmental stages (see text for details).

The homeobox gene tinman is expressed in the ventral mesoderm concomitant with *D-mef2* and subsequently becomes restricted in its expression to the visceral mesoderm (5, 6). At the same time, *D-mef2* becomes expressed at high levels in the somatic mesoderm. The normal expression pattern of *D-mef2* in the ventral mesoderm of tinman mutant embryos indicates that *D-mef2* is not regulated by tinman in these cells and suggests that these genes may be activated in parallel. Their coexpression in the committed mesoderm also raises the possibility that these gene products may collaborate to activate subordinate myogenic genes.

Somatic muscle cells are first detected late in the germ-band stage between 8 and 9 hr of development (2). nautilus transcripts first appear at a low level in unfused muscle precursor cells at about 7 hr of development, immediately prior to induction of myosin heavy chain, which marks the initiation of muscle differentiation (3, 4). After germ-band shortening, nautilus expression is restricted to precursor cells of the major groups of somatic muscles. However, nautilus expression is not detected in all somatic muscle cell precursors, which raises the question of whether it is required for the formation of the complete somatic musculature. Based on its early expression pattern, it appears that *D-mef2* may lie in a regulatory pathway "upstream" of nautilus. Whether *D-mef2* regulates nautilus expression remains to be determined. However, *D-mef2* alone cannot be sufficient for inducing nautilus because nautilus is expressed in only a subset of *D-mef2*-expressing cells. While genetic analyses will be required to fully define the role of *D-mef2* in regulating muscle gene expression, its pattern of expression in wild-type and mutant embryos suggests that *D-mef2* may function within a regulatory cascade as schematized in Fig. 5.

The conservation between *D-mef2* and the four vertebrate *MEF2* gene products within the MADS and MEF2 domains suggests that *D-mef2* represents an ancestral member of the *MEF2* family and that gene duplication events during evolution gave rise to the vertebrate *MEF2* multigene family. The conservation of the MADS domain, which mediates sequence-specific DNA binding, maintained the ability of these proteins to recognize the same DNA sequence and suggests that MEF2 proteins in *Drosophila* and in vertebrates regulate common sets of target genes.

The expression pattern of *D-mef2* is also reminiscent of that of the vertebrate *MEF2* genes. During mouse embryogenesis, members of the MEF2 family are initially detected within mesodermal cells destined to form the heart and within the myotomal compartment of the somites, from which skeletal muscle arises (12). In later stages of mouse embryogenesis, members of the *MEF2* gene family are expressed in a wide range of tissues, suggesting that with the expansion of the *MEF2* gene family during evolution, new members of the family may have acquired regulatory roles within cell types in addition to skeletal and cardiac muscle.

The restricted expression of MEF2 to the early skeletal and cardiac muscle lineages in *Drosophila* and vertebrates suggests that MEF2 may act at an early step in the genetic cascades leading to skeletal and cardiac muscle formation and that the functions of *MEF2* genes in regulating muscle-specific gene expression in these two striated muscle lineages have been conserved. Given the complexity of the vertebrate MEF2 family, elucidation of the precise roles of *MEF2* genes in the specification of skeletal and cardiac muscle cell fate may be facilitated by the existence of a single *MEF2* gene in *Drosophila*.

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